

TinyML-Based Real-Time Gait Phase Classification Using Low-Cost Sensor Fusion

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Abstract. Real-time gait phase classification constitutes a critical functional component in assistive robotic systems, wearable rehabilitation technologies, and human–robot interaction frameworks. However, practical deployment necessitates models that can function with low-cost sensing hardware, minimal latency, and constrained computational resources, while maintaining robust performance in the presence of noisy inertial measurements and substantially imbalanced distributions of gait phases. This paper introduces a TinyML-oriented processing pipeline for real-time gait phase classification based on low-cost inertial sensor fusion. The proposed framework integrates a wearable inertial measurement unit (IMU)-based prototype, signal preprocessing, sliding-window segmentation, temporal–spectral feature extraction, an XGBoost-based EdgeAI classifier, strategies for addressing class imbalance, and lightweight temporal post-processing. The system discriminates five distinct events of the gait cycle: Heel-Strike, Foot-Flat, Heel-Off, Toe-Off, and Swing. To mitigate overly optimistic performance estimates arising from the use of overlapping analysis windows, the evaluation is performed according to a file-wise leave-one-sequence-out cross-validation protocol, supplemented by an additional analysis on a held-out sequence. The proposed configuration achieved the best average LOSO composite score among the evaluated methods, reaching 0.6125, with a macro-F1 of 0.6480 and balanced accuracy of 0.6599. The ablation study showed that inertial sensor fusion was essential, while overly conservative rejection mechanisms reduced sensitivity to short transition phases such as Toe-Off. These results indicate that lightweight feature-based EdgeAI pipelines are a promising alternative for real-time gait phase classification with low-cost wearable sensors, while also highlighting the need for larger datasets, subject-independent validation, and deployment-specific profiling on embedded hardware.

Keywords: Gait Phase Classification · TinyML · EdgeAI · Inertial Sensor Fusion · Wearable Sensing · Assistive Robotics